Mod-1

ML:

↳ Machine learning is a branch of AI that focuses on developing models and algorithms that let computers learn from data without being explicitly programmed for every task.

→ ML teaches systems to think and understand like humans by learning from data.

ML

- Supervised → labelled data set
- Unsupervised → unlabelled data set
- Semi-supervised → small labelled dataset train ↳ unseen : improve accuracy.
- Reinforcement → learn through trial and error → decision making task.

* why do we need ML:

↳ Traditional programming require exact instructions and doesn't handle complex task like understanding image or language well. It can't efficiently process large amounts of data

ML solves this by learning from data without being explicitly programmed.

Supervised machine learning:

↳ It is a type of ML in which the model is trained using labeled data, meaning: each input has a corresponding correct output

"models learn from examples where answers are already known".

ex – Email filtering, house price prediction
→ handwritten to txt.

Supervise ML

Regression
↳ output is continuous (numbers)

ex:
↳ Predict temp
↳ Predict salary

classification
↳ output is categories (labels)

ex:
↳ spam / not spam
↳ pass / fail
↳ Yes / No, 0/1

Working

1) collect labelled data
2) Split the data
   (training) ↓ 80%
   (testing) ↓ 20%
3.) Train the model
   ↳ training data
4) Test the model
   ↳ new data, unseen data
5) Evaluate performance

Algorithms

i) Linear Regression
ii) Logistic Regression
iii) KNN
iv) Decision Tree
v) Naive Bayes

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Date ____
Page ____

Advantages

- ↳ high accuracy (if data is good)
- ↳ Easy to evaluate
- ↳ clear objective

Disadvantages

- ↳ Need labelled data (costly)
- ↳ overfitting problem
- ↳ Not good with unseen patterns.

Applications

i) email spam detection
ii) image classification
iii) medical diagnosis
iv) price prediction.

Unsupervised ML:

↳ unsupervised learning is a type of ML where the model is trained using unlabelled data, meaning there are no predefined output.

"model finds the pattern or structure in data without knowing the correct answers".

example: customer data: grouping similar customers
News articles: grouping by topic.

Working:

- find patterns
- group similar data
- detect hidden structure

→ finding relationship in data

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Types :

1.) Clustering.
↳ group similar data types together
ex: group similar customers on buying behaviour
Popular algo
↳ K-means
↳ Hierarchical clustering
↳ DBSCAN

2) Association:
↳ find relationship b/w variables

e.g. "People who buy bread also buy butter".

3.) Dimensionality Reduction :

↳ Remove number of features while keeping important info.
example: principle component analysis (PCA).

* How it works:
1.) collect unlabelled data
2) Apply algorithm.
3.) model finds patterns/groups
4.) Analyze results.
no " correct answer comparison.

Applications:
↳ customer segmentation
↳ Recommendation
↳ fraud detection
↳ market analysis.

Advantages:
↳ No need for labelled data
↳ useful for discovering hidden patterns
↳ works on raw data.

disadvantages:
↳ Hard to evaluate results
↳ less accurate than supervised
↳ interpretation is difficult

Semi-supervised learning: (SSL)

↳ SSL is a type of ML where the model is trained using a combination of labelled and unlabelled data.

"uses a small amount of labelled data and a large amount of unlabelled data."

* why do we need it:

In real life:

- labelled data = expensive & time consuming
- unlabelled data = easily avilabel

so, we mix both → better performance with less effort

How it works:

1) Train model using labelled data
2) use model to predict labels for unlabelled data.
3) combine both datasets
4) Retrain model for better accuracy.

* Applicatⁿ:	* Advantages:
↳ image classification	↳ Requires less labelled data
↳ speech recognition	↳ Improves accuracy compared to unsupervised
↳ Web-page classification	↳ cost effective
↳ medical data analysis.	

Disadvantages:

↳ wrong prediction can mislead model

↳ more complex than supervised

↳ Needs careful tuning.

Reinforcement learning:

↳ Reinforcement learning is a type of ML where an agent learns by interacting with an environment and improves it's performance by receiving reward or penalities.

"learning by trial and error using reward ~~and~~ or ~~punishments~~" penalties."

Agent → learner (AI model)
Environment → where agent acts
Action → what agent does
reward → feedback (+ or −)
State → current situation

How it works:
→ Agent observes environment
→ Takes an action
→ Receive reward/penalty
→ updates strategy
→ Repeat to maximize

Goal: Maximize total reward.

Types:

Positive reinforcement
↳ reward for good act^n

−ve reinforcement
↳ avoid bad outcome.

Advantages
↳ learn from real exp
↳ work well in dynamic environments
↳ improves over time

Disadvantages
↳ Takes time to train
↳ requires lots of data/interactions
↳ Can be complex.

Applications:
↳ Robotics
↳ self-driving cars
↳ Game AI
↳ Recommendation systems

Learning framework:

In ML datasets is divided into three parts

Training
→ Model learn
→ Model see input + correct output
→ learns patterns from data.

Validation
→ this use to tune the model
→ check performance during training.
→ Adjust parameters
→ Prevent overfitting.

Testing.
→ This is the final evaluation
→ Model is tested on new unseen data
→ Measure real performance
"Testing data is never used during training.

Performance evaluation:

↳ we use these to check how good our model is.

1.) Confusion matrix:

Structure

	Predicted +ve	Predicted -ve
Actual +ve	TP	FN
Actual -ve	FP	TN

TP → correctly predicted +ve
TN → correctly predicted -ve
FP → wrongly predicted +ve
FN → wrongly predicted -ve

2.) Accuracy:

$$Acc = \frac{TP + TN}{TP + FN + FP + TN}$$

→ how many predictions are correct overall.

3) Precision:

→ out of predicted +ve, how many are actually +ve. correct.

$$\text{Precision} = \frac{TP}{TP + FP}$$

4) Recall:

"out of actual positives, how many did we catch."

$$\text{Recall} = \frac{TP}{TP + FN}$$

5) F1 Score:

→ Balance b/w Precision & Recall

$$F1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

Module - 2

Regression:

* Linear Regression:

↳ Linear regression is a supervised learning algorithm used to predict a continuous output by finding a linear relationship b/w input and output variables.

"It fit's a straight line to predict value".

$$\boxed{Y = mx + b}$$

Y = Predict output
x = input
m = slope
b = intercept

How it works:

→ Take dataset (input + output)
→ Draw a line that best fits the data
→ Adjust line to minimize the error
→ use line to predict new values.

Goal:

→ Minimize the diff b/w:
- Actual value
- Predicted value

} → the difference is called error.

[Error = Actual - predicted]

Model tries to make this as small as possible.

Types

1.) Simple linear regression
↳ one input variable
↳ ex: Price vs size

2.) Multiple linear regression.
↳ multiple inputs
↳ Price vs size + location + rooms

Advantages:
↳ Simple and easy to understand
↳ fast to train
↳ works well for linear data

disadvantages:
↳ can't handle complex relationship
↳ Sensitive to outliers
↳ Assume linearity.

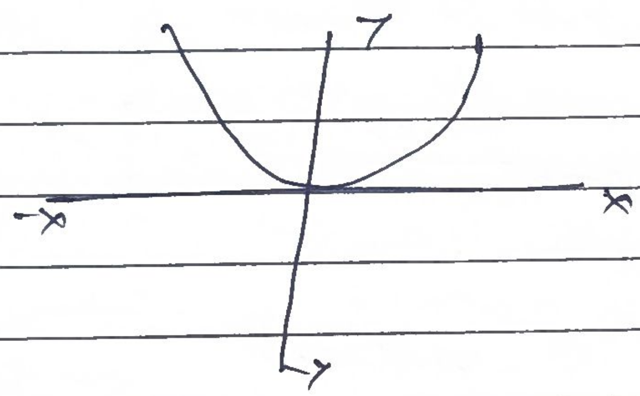
Applications:
↳ Price prediction
↳ sales forecasting
↳ Risk analysis.

Non-linear Regression:

↳ NLR is a type of regression where the relationship b/w input & output is not a straight line, but follow a curve or complex pattern.

"It models data using curves instead of straight line"

ex → population, temp, stock prices. → these follow curves, not straight line.



$$\boxed{y = ax^2 + bx + c}$$

Polynomial equation

* How it works:
1.) Take dataset
2.) choose a curve (polynomial, exponential etc.)
3.) fit curve to data
4.) minimize the error
5.) predict values.

Types
- Polynomial
- Exponential
- Logarithmic

* Advantages:
↳ Handles complex relationship
↳ more flexible than linear regression
→ better accuracy for real world data

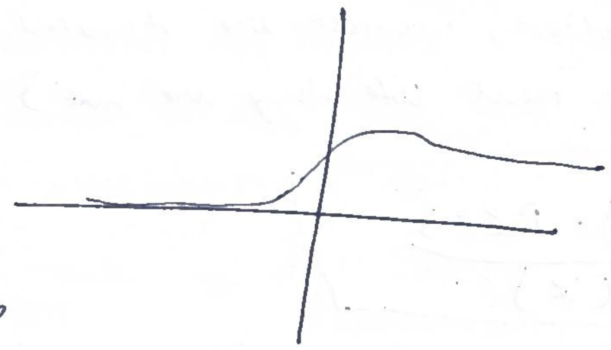
* Disadvantages:
↳ ~~more flexible than linear regression~~
↳ more complex
↳ Risk of overfitting
↳ harder to interpret.

Logistic Regression:

↳ Logistic regression is a supervised learning algorithm used for classification problems, where the output is categorical (usually 0 or 1).

"It predicts probability and converts it into a class label."

- if probablity > 0.5 → class = 1
- if probablity < 0.5 → class = 0



$$Y = \frac{1}{1+e^{-x}}$$

Sigmoid fun.

Sigmoid ?

↳ converts any value into range 0 to 1.

↳ Perfect for probablity.

How it works:

1) Take input data
2) Apply linear equation
3) Pass result through sigmoid funcⁿ.
4) Get probablity
5) convert into class (0 or 1).

Adv

→ simple & fast

→ works well for binary classification

→ outputs probablity.

disadv

→ only works well for simple problems

→ cannot handle complex boundaries

→ sensitive to outliers

Applican :

↳ spam detection

↳ Disease prediction

↳ fraud detection

Classification :-

Naive Bayes Algorithm:

↳ Naive bayes is a supervised ML algo based on Bayes' theorem. used for classification problem.

"It predicts class using probability and assumes features are independent".

Why Naive:

↳ Because it makes a naive (simple) assumption. All features are independent of each other.

- In spam detection, words are treated independently (even though in real life they are not)

$$\boxed{P(c|x) = \frac{P(x|c) \cdot P(c)}{P(x)}}$$

Where:

- $P(c|x)$ → Probability of class given input
- $P(x|c)$ → likelihood
- $P(c)$ → prior probability
- $P(x)$ → evidence.

Application:

↳ spam filtering
↳ sentiment analysis
↳ Document classification.

Naive ↓ assume everything independent.
bayes ↓ probability

→ generative model
→ logistic → discriminative model.

How it works:

1) calculate prior probability for each class.
2) calculate likelihood for features.
3) Apply bayes theorem
4) compute probability for each class.
5) choose class with highest probability.

TYPES

- Gaussian naive bayes
- Multinomial ...
- Bernoulli Naive bayes

Advantages:

↳ simple & fast
↳ works well with large data
↳ good for text classification

disadv:

→ Assume independence (not always true)
→ less accurate if feature are dependent.

K-NN:

↳ k-nearest Neighbors (KNN) is a supervised learning algorithm used for classification and regression, which classifies a data point based on the majority of its nearest neighbours.

"It Predicts output based on closest data points."

* basic idea:

↳ "Tell me who your neighbours are, and I'll tell you who you are".

- find nearest point
- check their class
- Assign majority class

ex → student pass/fail

↳ you have new student. check 5 nearest students:

- 3 are pass
- 2 are fail

} → Result : Pass

* How it works:

1.) choose value of k (no. of neighbours)

2.) calculate distance from new point to all data points.

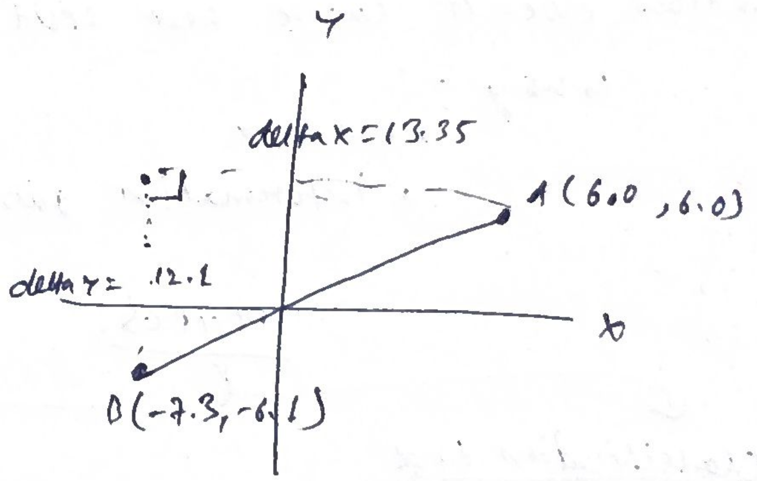
3.) select k nearest neighbours.

4.) count their classes

5.) Assign majority class.

$$\boxed{d = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}}$$

Euclidean distance formula.



- small k → sensitive to noise
- Large k → smoother but less accurate

→ usually choose odd number (avoid ties).

* Adv:

→ simple & easy

→ No training required

→ works for both classification and regression

* disadv:

→ slow for large datasets

→ ~~more~~ sensitive to irrelevant feature.

→ choosing 'k' is tricky.

* Applications:

→ Recommendation system.

→ pattern recognition.

→ image classification.

KNN → Nearest friends decide your category

Decision Tree:

↳ Decision tree is a supervised learning algorithm used for classification and regression, which makes decision using a tree-like structure of conditions.

"It splits data into branches based on decisions".

- Each node → question
- Each branch → answer
- Each leaf → final result

* How it works:

1.) Take dataset
2.) choose Best feature to split data
3.) Divide dataset into subsets
4.) Repeat spliting (Recursive)
5.) Stop when pure result is achieved.

↳ How does it choose "best split"?
  ↳ using :
    - Entropy
    - information gain.

high → Entropy → mixed data
low → Entropy → pure data

Types:

Classification tree
↳ output categories

Regression tree
↳ output numbers.

* Advantages

↳ easy to understand
↳ no need for scaling data
↳ works with both types (regression + classification)

* Disadv:

↳ can overfit
↳ sensitive to small changes
↳ complex tree become hard to interpret.

* Application:

↳ medical diagnosis
↳ customer decision analysis.
↳ Risk assessment.